

# **Project Proposal**

**Smart Plug (Voltiva 1.0)**

**Group No: 07**

## **❖ What have we done?**

In the previous semester, the project focused on developing a basic IoT-based energy monitoring system capable of measuring voltage, current, and power usage of household appliances. Using components like the ACS712 current sensor, ZMPT101B voltage sensor, and ESP32 microcontroller, the system collected real-time data and transmitted it wirelessly via Wi-Fi. A simple web interface displayed key energy parameters.

Simulations and breadboard-based hardware tests were conducted to verify the design and functionality. However, the system had several limitations, including a lack of data storage, no advanced analytics, and a basic user interface. These limitations were identified for future improvements, forming the foundation for the current semester's upgrades.

## **❖ What are we doing?**

The IoT Smart Energy Meter utilizes an ESP32 microcontroller integrated with ACS712 current and ZMPT101B voltage sensors to measure electrical parameters. The system calculates real power, apparent power, and power factor, transmitting this data via Wi-Fi to the Blynk 2.0 platform for real-time monitoring on web and mobile interfaces. Additionally, it incorporates a Telegram bot to send energy consumption reports and billing information to users.

Building upon this foundation, our current semester's upgrades focus on enhancing functionality, user experience, and system scalability. Key improvements include:

### **1. Telegram Bot Integration:**

- Replaced the new project's basic alerts with Telegram bot notifications (improvement over SMS/GSM modules).

### **2. Web Application Improvement:**

- The new project's basic monitoring was expanded into a cross-platform app "Blynk 2.0" (JS + HTML + CSS), avoiding previous problems for better usability.

### **3. Adding Mobile Application:**

- Creating a mobile application, in addition to the web application, using "Blynk 2.0" for ease of use for users.

### **4. Calibration:**

- Explicitly prioritized sensor calibration (a noted gap in our past and present projects).

### **5. AC/DC Separation:**

- Adding clear isolation of high/low-voltage components (aligned with the new project's safety focus).

These advancements aim to transform the basic energy monitoring system into a comprehensive, intelligent energy management solution, aligning with modern smart home and industrial requirements.